



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc. EXAMINATION

COLLEGE: SCIENCE & TECHNOLOGY

DEPARTMENT: COMPUTER & INFORMATION SCIENCES

SESSION: 2024/2025

COURSE CODE: CSC416

COURSE TITLE: DISCRETE STRUCTURES

INSTRUCTION: ATTEMPT ANY FOUR (4) QUESTIONS

SEMESTER: ALPHA

CREDIT UNIT: THREE (3)

TIME: 3 HOURS

Question One

a. State the converse, contrapositive, and inverse of each of these conditional statements.

i. If it snows tonight, then I will stay at home.

ii. I go to the beach whenever it is a sunny summer day.

iii. When I stay up late, it is necessary that I sleep until noon.

4.5 marks

b. Construct a truth table for each of these compound propositions.

i. $p \leftrightarrow \neg p$

ii. $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

iii. $(p \oplus q) \vee (p \oplus \neg q)$

iv. $((p \rightarrow q) \rightarrow r) \rightarrow s$

4 marks

c. Consider the conditional statement:

"The home team wins whenever it is raining."

Let p represent: *"It is raining."*

Let q represent: *"The home team wins."*

i. Using this, find the following: the contrapositive, the converse, and the inverse of the statement.

3 marks

ii. Prove that only the contrapositive of a conditional statement $p \rightarrow q$ has the same truth value as the original conditional statement. Use the statement from c(i) in your explanation.

6 marks



Question Two

- a. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.
- A student in your class has a cat, a dog, and a ferret.
 - All students in your class have a cat, a dog, or a ferret.
 - Some student in your class has a cat and a ferret, but not a dog.
 - No student in your class has a cat, a dog, and a ferret.
 - For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.
- 7.5 marks
- bi. Let $f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c and d are constants. Determine necessary and sufficient conditions on the constants a, b, c and d so that $f \circ g = g \circ f$. 2 marks
- ii. If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer. 2 marks
- iii. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from R to R . 2 marks
- c. Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression involving predicates, and quantifiers with a domain consisting of all people, and logical connectives. 4 marks

Question Three

- a. Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same. 3.5 marks
- b. Write a C++ program to generate all possible subsets (also known as the power set) of a given set of integers.

For example, if the user enters the set $\{1, 2, 3\}$, the output should include:

{ }
 { 1 }
 { 2 }
 { 1, 2 }
 { 3 }
 { 1, 3 }
 { 2, 3 }
 { 1, 2, 3 }

6 marks

- c. Use set builder notation and logical equivalences to establish the first De Morgan law $A \cap \bar{B} = \bar{A} \cup \bar{B}$. 3 marks
- d. The bit string for the set $\{1, 3, 5, 7, 9\}$ with universal set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ is 10 1010 1010.
- What is the bit string for the complement of this set? 2.5 marks

let $C = A \cap B$
 $C = \{x \mid x \in A \wedge x \in B\}$

- ii. The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are $11\ 1110\ 0000$ and $10\ 1010\ 1010$, respectively. Use bit strings to find the union and intersection of these sets.

2.5 marks

Question Four

- a. Determine the truth value of each of these statements if the domain consists of all real numbers.

i. $\exists x(x^3 = -1)$ ii. $\exists x(x^4 < x^2)$ iii. $\forall x((-x)^2 = x^2)$ iv. $\forall x(2x > x)$

4 marks

- b. Consider the following relations on $\{1, 2, 3, 4\}$:

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$

$R2 = \{(1, 1), (1, 2), (2, 1)\}$

$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$

$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$

$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$

$R6 = \{(3, 4)\}$

- Which of these relations are reflexive?
- Which of these relations are symmetric?
- Which of these relations are antisymmetric?
- Which of these relations are transitive?
- Which of these relations are reflexive?

7.5 marks

- c. Write a C++ function called `isPerfectSquare` that uses a binary search algorithm to verify if a given integer num is a perfect square. The function should:

- Narrow down the search range $[low, high]$ by repeatedly dividing it into two parts (midpoint).
- Calculate the square of the midpoint value.
- Check if the square of the midpoint is equal to num. If so, return the midpoint as the square root of num.
- If the square of the midpoint is greater than or less than num, adjust the search range accordingly.
- If the search range is exhausted and no midpoint satisfies the condition, return -1.

Write a main function to:

- Prompt the user to enter an integer `xx`.
- Call `isPerfectSquare(x)`.
- Display whether `xx` is a perfect square. If `xx` is a perfect square, also display its integer square root.

6 marks

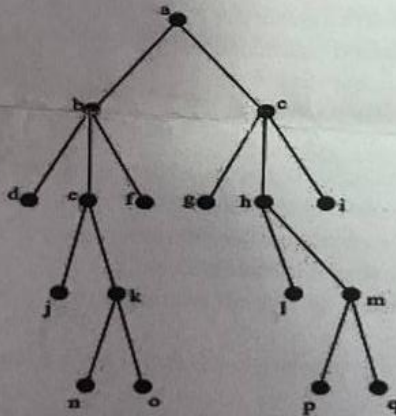
Question Five

A graph designed by a 400-level Graphics student consists of 24 vertices and 30 edges. The graph has five vertices of degree 4, seven pendant vertices, and seven vertices of degree 2. All other vertices have a degree of either 3 or 4. After completing the design, the student seeks your assistance as a computer science student to help solve the following questions for further review:

- How many vertices of degree 4 are there in the given graph? 4 marks
- What is the total number of vertices of degree 4 in the corresponding complete graph? 1 mark
- Provide a proof to determine whether the Handshaking Theorem holds for this scenario. 3 marks
- Write a C++ code snippet to compute the remaining number of vertices that have either degree 3 or 4. 3 marks
- Demonstrate how the Handshaking Theorem can be applied in Data Communication. 2 marks
- Prove that if G is a graph with n vertices and n edges, and no vertex has a degree of 0 or 1, then the degree of every vertex in G is 2. 3 marks
- Explain what it means for two graphs G and H to be isomorphic. 1.5 marks

Question Six

- Consider the following tree



- List the vertices in the order that you would visit them if you traversed the tree in preorder. 2 marks
 - List the vertices in the order that you would visit them if you traversed the tree in inorder. 2 marks
 - List the vertices in the order that you would visit them if you traversed the tree in postorder. 2 marks
- Consider the expression: $(3 - 4 \cdot 2)^4 - (5 + (5 - 2)^4)$
 - Draw the rooted tree representing this computation 2 marks
 - Write this expression in prefix notation 2 marks
 - Write this expression in postfix notation. 2 marks
 - A quinary tree has the height of 3. Find the total number of leaves present on it. 2.5 marks
 - Prove that a quinary tree is an m -ary tree of height 3 has at most m^3 leaves. 3 marks